

2016 Summer Qual

III. $\Sigma = LL^T$: Cholesky decomposition, L : full column rank

$$y = Lz \sim N_n(0, \Sigma), \quad z \sim N(0, I)$$

$$y^T A y = z^T \underbrace{L^T A L}_B z$$

$$B^T = (L^T A L)^T = L^T A L = B : \text{symmetric}$$

$$L^T A L L^T A L L = L^T A L L^T$$

Since L is full column rank, $\exists (L^T L)^{-1}$

$$\cancel{(L^T L)^{-1} L^T L} L^T A L L^T A L L = \cancel{(L^T L)^{-1} L^T L} L^T A L L$$

$$\Rightarrow L^T A L L^T A L L = L^T A L L$$

$$\Rightarrow L^T A L L^T A L L L^T L \cancel{(L^T L)^{-1}} = L^T A L L L^T L \cancel{(L^T L)^{-1}}$$

$$\Rightarrow L^T A L L^T A L = L^T A L$$

$$\Rightarrow L^T A L : \text{idempotent}$$

$$\text{rank}(L^T A L) = \text{tr}(L^T A L) = \text{tr}(A L L^T) = \text{tr}(A \Sigma) = r$$

$$z^T B z = y^T A y \sim \chi_r^2$$

3.

$$\hat{\beta}^* = (X_1^T X_1)^{-1} X_1^T Y \Rightarrow E[\hat{\beta}^*] = (X_1^T X_1)^{-1} X_1^T [X_1 \beta_1 + X_2 \beta_2] \\ = \beta_1 + (X_1^T X_1)^{-1} X_1^T X_2 \beta_2$$

$$\text{bias of } \hat{\beta}^* = (X_1^T X_1)^{-1} X_1^T X_2 \beta_2$$

$$\text{var}(\hat{\beta}_1^*) = \sigma^2 \mathbf{d}_1^T (X_1^T X_1)^{-1} \mathbf{d}_1$$

$$\text{var}(\hat{\beta}_0) = \sigma^2 \mathbf{d}_0^T (X^T X)^{-1} \mathbf{d}_0$$

$$= (\mathbf{d}_1^T \mathbf{d}_2^T) \begin{pmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{d}_1 \\ \mathbf{d}_2 \end{pmatrix}$$

$$= (\mathbf{d}_1^T \mathbf{d}_2^T) \begin{pmatrix} Q^{11} & Q^{12} \\ Q^{21} & Q^{22} \end{pmatrix} \begin{pmatrix} \mathbf{d}_1 \\ \mathbf{d}_2 \end{pmatrix}$$

$$Q^{11} = (X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 (X_2^T (I - H_{X_1}) X_2)^{-1} X_2^T X_1 (X_1^T X_1)^{-1}$$

$$Q^{12} = - (X_1^T X_1)^{-1} X_1^T X_2 [X_2^T (I - H_{X_1}) X_2]^{-1}$$

$$Q^{21} = - [X_2^T (I - H_{X_1}) X_2]^{-1} X_2^T X_1 (X_1^T X_1)^{-1}$$

$$Q^{22} = [X_2^T (I - H_{X_1}) X_2]^{-1}$$

$$\text{var}(\hat{\beta}_0) - \text{var}(\hat{\beta}_1^*)$$

$$= (\mathbf{d}_2 - X_2^T X_1 (X_1^T X_1)^{-1} \mathbf{d}_1)^T \underbrace{Q^{22}}_{\text{p.s.d.}} (\mathbf{d}_2 - X_2^T X_1 (X_1^T X_1)^{-1} \mathbf{d}_1)$$

≥ 0

□

4. For $\text{cov}(e) = \frac{\sigma^2}{2}(\mathbf{I} + \mathbf{J})$,

$$\mathbf{y}^* = \sqrt{2}(\mathbf{I} + \mathbf{J})^{-1/2} \mathbf{y}$$

$$\mathbf{X}_c^* = \sqrt{2}(\mathbf{I} + \mathbf{J})^{-1/2} \mathbf{X}_c$$

$$\mathbf{1}^* = \sqrt{2}(\mathbf{I} + \mathbf{J})^{-1/2} \mathbf{1}$$

$$\frac{\partial \|\mathbf{y}^* - \mathbf{1}^* \alpha - \mathbf{X}_c^* \beta_c\|^2}{\partial \alpha} = 0$$

$$\Rightarrow -2 \mathbf{1}^{*T} (\mathbf{y}^* - \mathbf{1}^* \alpha - \mathbf{X}_c^* \beta_c) = 0$$

$$\mathbf{1}^{*T} \mathbf{y}^* = \mathbf{1}^{*T} \mathbf{1}^* \alpha$$

Using Sherman - Morrison Lemma

$$\mathbf{1}^{*T} \mathbf{1}^* = 2 \mathbf{1}^T (\mathbf{I} + \mathbf{J})^{-1} \mathbf{1}$$

$$= 2 \mathbf{1}^T (\mathbf{I} + \mathbf{1} \mathbf{1}^T)^{-1} \mathbf{1}$$

$$= 2 \mathbf{1}^T \left(\mathbf{I} - \frac{1}{1+n} \mathbf{J} \right) \mathbf{1}$$

$$= \frac{2n}{n+1}$$

$$\hat{\alpha} = \frac{2}{n+1} \sum y_i \cdot \frac{n+1}{2n} = \bar{y}$$

$$\frac{\partial \|\mathbf{y}^* - \mathbf{1}^* \alpha - \mathbf{X}_c^* \beta_c\|^2}{\partial \beta_c} = 0$$

$$\Rightarrow -2X_C^T (Y^* - 1^* \alpha - X_C^* \beta_C) = 0$$

$$\hat{\beta}_C = (X_C^{*T} X_C^*)^{-1} X_C^{*T} Y^*$$

$$= \frac{n+1}{2n} \cdot \frac{2n}{n+1} (X_C^T X_C)^{-1} X_C^T Y$$

$$= (X_C^T X_C)^{-1} X_C^T Y$$

$\therefore$ same.



[5]

$$y_{ij} = \mu + a_i + e_{ij}, \quad a_i: \text{missing data}$$

$$a_i \stackrel{\text{iid}}{\sim} N(0, \sigma_a^2), \quad e_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad a_i \perp e_{ij}$$

$$y_{ij} | a_i \sim N(\mu + a_i, \sigma^2), \quad a_i \sim N(0, \sigma_a^2)$$

$$\begin{aligned} \ell(\theta) = & -\frac{mn}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m \sum_{j=1}^n (y_{ij} - \mu - a_i)^2 \\ & -\frac{m}{2} \log \sigma_a^2 - \frac{1}{2\sigma_a^2} \sum_{i=1}^m a_i^2 \quad (\text{상수항 제외}) \end{aligned}$$

$$\theta^{(k)} := (\mu^{(k)}, \sigma_a^{2(k)}, \sigma^{2(k)})$$

$$\text{Need: } \mathbb{E}(a_i | y, \theta^{(k)}), \quad \mathbb{E}(a_i^2 | y, \theta^{(k)})$$

$$\bar{y}_i := \frac{1}{n} \sum_{j=1}^n y_{ij}$$

$$\bar{y}_i | a_i \sim N(\mu + a_i, \frac{\sigma^2}{n})$$

$$\& a_i \sim N(0, \sigma_a^2)$$

$$\mathbb{E}(a_i | y, \theta^{(k)}) = \frac{\sigma_a^{2(k)}}{\sigma_a^{2(k)} + \frac{\sigma^{2(k)}}{n}} (\bar{y}_i - \mu^{(k)})$$

$$=: m_i^{(k)}$$

$$\mathbb{E}[a_i^2 | y, \theta^{(k)}] = \text{Var}(a_i | y, \theta^{(k)}) + \{m_i^{(k)}\}^2$$

$$\text{Var}(a_i | y, \theta^{(k)}) = \left(\frac{1}{\sigma_a^{2(k)}} + \frac{n}{\sigma^{2(k)}} \right)^{-1} = \frac{\sigma_a^{2(k)} \sigma^{2(k)}}{\sigma_a^{2(k)} + n\sigma^{2(k)}} =: v_i^{(k)}$$

$$\Rightarrow \mathbb{E}[a_i^2 | y, \theta^{(k)}] = v_i^{(k)} + \{m_i^{(k)}\}^2$$

E-step

$$Q(\theta|\theta^{(k)}) = -\frac{mn}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m \sum_{j=1}^n \mathbb{E}^{(k)} [(y_{ij} - \mu - a_i)^2]$$

$$-\frac{m}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m \mathbb{E}^{(k)} (a_i^2) \quad (\text{상수항 제외})$$

Here, $\mathbb{E}^{(k)} [(y_{ij} - \mu - a_i)^2]$

$$= \mathbb{E}^{(k)} [(y_{ij} - \mu)^2 - 2a_i(y_{ij} - \mu) + a_i^2]$$

$$= (y_{ij} - \mu)^2 - 2(y_{ij} - \mu) \underbrace{\mathbb{E}^{(k)}(a_i)}_{m_i^{(k)}} + \underbrace{\mathbb{E}^{(k)}(a_i^2)}_{v_i^{(k)} + (m_i^{(k)})^2}$$

$$= (y_{ij} - \mu - m_i^{(k)})^2 + v_i^{(k)}$$

$$Q(\theta|\theta^{(k)}) = -\frac{mn}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m \sum_{j=1}^n [(y_{ij} - \mu - m_i^{(k)})^2 + v_i^{(k)}]$$

$$-\frac{m}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m [v_i^{(k)} + (m_i^{(k)})^2] \quad (\text{상수항 제외})$$

M-step

① update μ

$$\frac{\partial Q}{\partial \mu} \Rightarrow \frac{\partial}{\partial \mu} \sum_{i=1}^m \sum_{j=1}^n (y_{ij} - \mu - m_i^{(k)})^2$$

$$\Rightarrow \sum_{i=1}^m \sum_{j=1}^n (y_{ij} - \mu - m_i^{(k)}) = 0$$

$$\hat{\mu}^{(k+1)} = \bar{y}_{..} - \frac{1}{m} \sum_{i=1}^m m_i^{(k)}$$

② update for σ^2

$$-\frac{m}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^m [v_i^{(k)} + (m_i^{(k)})^2]$$

$$\frac{\partial Q}{\partial \sigma^2} \Rightarrow \hat{\sigma}_\sigma^{2(k+1)} = \frac{1}{m} \sum_{i=1}^m [v_i^{(k)} + (m_i^{(k)})^2]$$

③ update for σ^2

Similar to σ_a^2 ,

$$\sigma^2(t+1) = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [(g_{ij} - \mu(t+1) - m_j(t))^2 + v_j(t)]$$

